

Challenges of Applying Ethnography to Study Software Practices

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ABSTRACT

Ethnography is about the adoption of a cultural lens to observe and interpret events, actions, and behaviors, ensuring that they are placed in a relevant and meaningful context. Using this approach, it is possible to capture and analyze software development practices. Our aims are to illustrate the use of an ethnographic approach in a case study of agile software development adoption, to discuss the methodological challenges involved, and to provide support to others who conduct ethnographic studies of software practice. An ethnographic case study was conducted, employing participant observation, interviews, and document analysis. Difficulties and decisions were recorded and compared with those encountered in the literature. Finally, key challenges and guidelines to tackle them were discussed and documented. We identified five key challenges of applying ethnography to the study of software practices: (a) working in collaboration with and having something to offer to the participating company; (b) the insider/outsider dynamic of participant observation; (c) the balance between participant listening and participant observation; (d) the researcher's relationship with the participants; and (e) the rigor in qualitative work that involves the dilemma of the contextualization to be sufficiently broad and detailed. This study shows that ethnographic methods are indispensable when trying to understand software practice, and that the fundamental challenge for the researcher is to balance the role of participant observer with rigorous fieldwork.

Categories and Subject Descriptors

D.2 [Software Engineering]

General Terms

Management, Measurement, Experimentation, Human Factors.

Keywords

Qualitative Methods; Collaborative Ethnographic Research; Participant Observation.

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1. INTRODUCTION

Ethnography is about telling a credible, rigorous, and authentic story, giving voice to people in their local context. This story is told through the eyes of people as they pursue their daily lives in their own communities [7]. Typically, the ethnographic story relies on verbatim quotations and descriptions of scenarios that allow an *inside* perspective of the context of the people under observation. Ethnography derives from traditional anthropology where time in the field is needed to get a comprehensive description and understanding of a setting, group, or culture under study [7].

Ethnographic research methods have played a substantial role in sociological research during the last half-century and are now used in many disciplines, especially those that involve social and human factors [3]. They have become widely used in the study of information systems and can be a highly useful methodology for addressing a range of research questions related to SE practices [2][10][15][17][21][22].

In particular, ethnographic methods are helpful in generating rich and detailed accounts of software project teams, their interactions with project stakeholders, and their approaches for delivering products, as well as in-depth accounts of their experiences. Ethnography is a way of getting close to the reality of a social phenomenon, of a software team's interactions, communications, and relationships. Additionally, since it aims to generate holistic social accounts, ethnographic research can identify, explore, and link social phenomena, which on the surface, seem to have little connection with each other.

However, applying an ethnographic approach to SE can be both challenging and demanding. The problems and objects of study in SE, by their very specific nature, require empirical methods suited to specific contexts. Certain phenomena, for example, require a specialist understanding of the concepts under study and extensive data collection that lasts for months and years, rather than days and weeks. Also, the ethnographic researcher needs to be aware of the effects on the behavior of the people being studied [12].

The aim of this paper is to present the methodological challenges of applying ethnography to a case study aimed at characterizing an agile project team's belief system and its origins, sources, and impacts on the team's software development practices [15]. In particular, we focus on the social interactions, communications, and relationships that arise as an intrinsic part of adopting agile

software development practices. For that, we applied an ethnographic approach, employing participant observation, interviews, and document analysis. The main idea of our approach involved performing ethnography with its holistic and contextual vision, including some characteristics of action research, such as its collaborative and reflexive approach; used to recommend how to improve the studied practices to the company.

In particular, we address: (a) the advantages and challenges associated with working in collaboration with, and having results to offer to, the participating company (b) the insider/outsider dynamic of participant observation; (c) the balance between participant listening and participant observation; (d) the researcher's relationship with the participants; and (e) the rigor in qualitative work that involves the dilemma of the contextualization to be sufficiently broad and detailed. We relate our findings to relevant ethnographic theory, e.g., [3][7][8][11][12][14][24][25] to contribute to a better understanding on how to apply ethnography to study software practices.

The rest of the paper is organized as follows: Section 2 introduces ethnography as a research method. Section 3 describes how we conducted the ethnographic case study. Section 4 discusses the challenges of applying ethnography as a research approach, implications for research and practice, and limitations. Conclusions and opportunities for further work are presented in Section 5.

2. BACKGROUND

Social anthropologists, to aid in the understanding of different cultures and environments, originally founded ethnographic research methods. Ethnography is about living in and immersing in the culture of the researched group for an extended period of time. It is about participating in social relations, seeking to understand actions within the context of an observed setting, and how people act and make sense of their environment. A key purpose of ethnography is to provide a detailed, in-depth description of everyday life and practice [8].

Ethnographic methods seek to understand practice in its own terms, while trying to minimize the impact of the researchers' backgrounds, prejudices, and assumptions. Ethnographers adopt a cultural lens to interpret observed events, actions, and behaviors, ensuring that they are placed in a culturally relevant and meaningful context. It is also important to participate in social relations and interactions to gather data by an active attitude in an observed setting.

Fieldwork is the most characteristic element of the ethnographic method. Typically, and concurrently with data collection, ethnographers devote a large amount of their fieldwork time to formally analyze and reanalyze their data [7]. While still in the field, interim reports, which are the beginnings of the ethnographically informed final report, are produced to enable feedback on the data-gathering cycle. Narratives can be made and revised to take into account the ethnographer's evolving knowledge.

In this context, the primary research technique is *participant observation* [6], which involves direct observation, participation in the life of the observed group, and collective discussions through an intensive involvement with people in their natural environment, usually over an extended period of time. It is a three-stage process in which the researcher first, gains access to a

particular community. Second, lives and/or works among the people under study in order to grasp their world views and ways of life, and third, returns to make sense of the collected data through writing up an account of the community's culture [3]. The process of immersion may be unsystematic in the beginning, being somewhat uncontrolled and casual. However, even in the early stages of fieldwork, the ethnographer searches out experiences and events as they come to his/her attention. Thus, the participant observation technique becomes more refined, as the researcher understands more about the culture.

The acquisition of knowledge in ethnographic research is a cyclical process. It begins with a panoramic view of the community, closes in to a fine focus on details, and then goes back to the larger picture again, but this time with detailed information. The focus narrows and widens repeatedly during the process, because only by both deep diving and swimming on the surface can the ethnographer portray the cultural landscape in detail rich enough for others to comprehend and appreciate [7].

The traditional principles of participant observation and ethnographic fieldwork require a full immersion of the researcher in the chosen field of study, learning the day-to-day and extraordinary essence of social and cultural life by *being there* [18]. Ideally, the researcher does not disturb or change the natural setting.

For most anthropologists, ethnography involves actually living in the communities of the studied people, participating in their activities, interviewing them, drawing maps of the context, and collecting artifacts [9]. Critics of the approach are concerned about researchers getting over-involved with the people and environment under study, perhaps disturbing and changing the natural setting [18]. By contrast, some sociological ethnographers do not actually live with the people they study or spend most of their time with them. Instead, they focus on what happens in a particular institution when in operation; their participant observation is part-time, but sufficient to complete the research. In any case, ethnography must have a holistic view of the culture or group under study.

One can build a blueprint or roadmap to help the ethnographer conceptualize the steps to build knowledge and understanding. The goal is to find an effective problem-solving sequence [6]. The fieldwork approach shapes the design of all ethnographic work. The ethnographer typically begins with a survey period to learn the basics: language, history, structure, and function of the culture under study. After that it is possible to draw social and conceptual boundaries to identify or confirm themes, subjects, problems, and questions of interest to compose the research design. The next step is to start the fieldwork to observe, interview, read, ask insightful questions, and to write down what is seen and heard. Life histories of articulate individuals can be especially illuminating. The ethnographer must then crosscheck, compare, and triangulate the findings [3].

Proper organization from the very beginning of the work is a good practice. Keeping data organized and at hand allows the ethnographer to make interim reports and anticipate testing hypothesis throughout the investigation. Analysis and report writing are more efficient when the data is structured during fieldwork. During the analysis, it is possible to examine hypothesis and perceptions to construct or refine a conceptual framework about the studied group or culture [7].

Table 1. Ethnography vs. Action Research

	ETHNOGRAPHY	ACTION RESEARCH
Goal	Go Native In-Depth Understanding of Culture Generate Rich and Detailed Social Account	Process Improving Knowledge Advancing Problem-Solving
Focus	Culture and Values	Process and Practices
Main Technique	Participant Observation	No specific technique
Fieldwork	Holistic Comparative Contextual	Essentially Collaborative Reflexive Problem-Solving Driven
Final Product	In-Depth Description of Group's Life	Improved practice New knowledge

Some ethnographers spend as much time formally analyzing and reanalyzing their data and writing their ethnographies as they do conducting fieldwork. The final stage of analysis is the most creative step of ethnographic research, because the synthesis of ideas and field information often lead to useful insights through allowing the mind to wander and consider unusual combinations of thoughts. However, the researcher must always check whether the data will support the findings or invalidate them.

Research proposals, field notes, memos, web sites, interim reports, final reports, articles, and books are the tangible products of ethnography. The written work can be shared with participants, to verify its accuracy, and with colleagues for review and consideration. If the ethnographer is a collaborative researcher, he or she might share documents with community members, who can edit and co-write the ethnographic findings.

The decision of leaving the field should be based on more than one reason [6]. The best reason is the belief that enough data has been gathered to say something significant about it, but sometimes the proximity of a deadline is the determining reason. Different researchers require different levels of confidence about research findings and conclusions, but when the general picture reaffirms itself over and over again, it is probably time to leave the field and continue the job from outside.

After leaving the field, the ethnographic work continues with the final stages of analysis in which field notes, interim reports, papers, and so on are used to draw an overall picture of how the practice system works and which data will sustain or break the findings. Thus, ethnography is both a research method and a product, typically in the form of a written text.

Nowadays, a more collaborative perspective to ethnography is commonly required in organizational contexts. In contrast to traditional ethnography, contemporary collaborative ethnographic research has applied strategies, such as participatory action research [2][6][11][20][23], that aim not only to analyze but also to improve work practices. Dittrich et al., for example, adopted an ethnographically inspired action research approach, which combines qualitative social science fieldwork, with problem-oriented method, technique, and process improvement [5]. However, it is noteworthy that *ethnography* and *action research* are two different approaches (as showed in Table1). Ethnography focuses on the culture and values of a target group with the aim of generating rich and detailed accounts to produce an in-depth description of the group's life, while action research focuses on processes and practices and how to improve them in a problem-solving paradigm [11].

3. CONDUCTING AN ETHNOGRAPHIC CASE STUDY

We have been running a long-term case study, involving software development projects, which has gone through a main cycle of 18 months. This study is now undergoing another cycle, lasting approximately 12 months, with the aim of characterizing an agile project team's belief system and to investigate its origins, sources, and impacts on the team's software development practices [15]. In particular, we have been focusing on the social interactions, communications, and relationships that arise as an intrinsic part of adopting agile software development practices. To do this, we have been applying an ethnographic approach, immersing ourselves in the day-to-day project activities to delve deeper into the complexity of team's belief system and uncover meanings of behaviors, actions and events.

The case study was set in a medium sized software company¹ in Brazil, in which projects were applying, for the first time, agile methods based on Scrum². The company provides software development and evolution services as well as information technology infrastructure management services for customers of both the public and private sectors.

We adopted an approach that aimed explicitly to help the company to improve the work practices under study, getting some characteristics of action research, but mainly based on more modern approaches of ethnography [2][8][11][12]. In our context, this participatory approach was required, so we worked collaboratively to support their process improvement initiative.

The first author of the paper performed the interviews, observations, and document analysis according to [13][18][19]. It is important to note that the first author is an employee of the company in leave for PhD study. Therefore, there was a pre-existent common language between the interviewer and the participants of the study.

The particular project studied involved a system for electrical energy management and lasted for one year. The project team followed a well-defined process for software development and used a project management application for stories and bug tracking. They also kept detailed records of the project status reports and the backlog of stories and changes to the software. Other important documents, including quantitative measures, were used for project tracking and monitoring (Table 2).

Table 2. Project Profile

Attribute	PGE Project
Team Size	5
Team Members	2 full time developers, 2 part time developers, 1 scrum master, 1 product owner + project manager
Language	JAVA
Tool Usage	Open Source Software
Development Methodology	SCRUM + PMI Approaches
Non-Functional Requirements	Reliability, Performance, Continuity, Availability
Reuse	High
Stability Requirements	Medium stability
Staff Turn-over	Considered high by the project manager
Interviews	2 with product owner + project manager 2 with scrum master

¹ SOLUTIS Tecnologias. <http://www.solutis.com.br/>

² Scrum approach. <http://www.Scrum.org/>

3.1 Interviews

After a literature review and evaluation of research objectives and questions, we opted for an interview-based qualitative data collection approach called War Story [13]. War Story questionnaires usually have warm-up, past experience, lessons learned, and reaction questions. We used an iterative approach in which we defined a questionnaire (see Appendix A), used it for a set of interviews, analyzed the data, and improved the process for the next round of interviews.

Warm-up questions aimed to put the interviewees in the right mindset, to focus them on the interview subject. Our version of the questionnaire had two warm-up questions that asked about the participants' background and experience, and the main challenges of the current software project.

Past experience questions intended to investigate how living experiences could influence the participants' current behavior, trying to cover the main aspects of a software project reality. Our questionnaire had four of these questions. These questions are about the respondents' past experiences, involving software process, communication plan, software engineering technical practices, and project monitoring.

Lessons learned questions tried to capture the beliefs that emerged or evolved from the project experiences. To keep the questionnaire simple and balanced for the research kick-off, we had three war story questions about lessons learned. They aimed to understand the basis of how the beliefs, personal values, and the paradigms of the participants came to be and also how they determined the participants' choices and actions during the project.

Reaction questions asked about the participants' reactions and personal opinions on the impacts of the use of technologies, methods, and process on the software project. The questionnaire had two reaction questions that directly asked about the impacts of new technologies and unexpected effects of known and new technologies on projects.

Besides the typical war story questions, we added a few questions to identify beliefs related to practices that affected project *productivity, quality, time, and cost*. Those attributes are usually quantitatively measured in software organizations, so they helped to link quantitative and qualitative findings (*Metric-based Questions*). We had four of them in the questionnaire.

The first author asked the respondents to retell and revive specific and directed stories that illustrated the experiences we were trying to capture. The resulting data contained a considerable amount of contextual information, which enabled connections between different, but related, stories.

3.2 Observation and Document Analysis

Field notes and recordings of relevant interviews on completion of observations in meetings provided rich insight into social relations, events, and practices. We established specific goals for our observation task and some guided questions to drive the researchers' work (Appendix B). During observations, we immersed ourselves in the setting, using an informal approach, which allowed us to probe emerging issues about unusual events in a naturalistic manner.

Additionally, we collected organizational documents and attended team meetings. We compared our field notes from interviews and observations with the practices and processes laid down in documents and information about tasks. We also used document

analysis to map the company and project contexts (see Appendix C). By triangulating our data sources and our instruments, we addressed issues of validity and obtained comprehensive insights into the application of ethnographic methods.

3.3 Study Analysis and Results

The first author conducted four interviews with two people in the front lines of the project that introduced agile programming recently. Each person was interviewed twice, once at the beginning and once at the end of the project, six months later. These four interviews were complemented by meeting observations and document analysis. The first author attended and recorded 40 meetings involving the interviewed participants and all the team members.

All field notes, including transcripts and recordings of meetings and interviews, were categorized, tabulated, and then analyzed via cycles of pattern coding, which labeled keywords, phrases, and paragraphs to group and summarize segments of text into classes of findings.

We analyzed the relevant events of past and current projects that might influence team beliefs or practices, including the SCRUM methodology training. Considering these scenarios, we captured and classified the beliefs found and described their main attributes: origin, context, meaning, impact, participant, and frequency.

We represented our main findings in a mind map [15], which is a diagram used to represent concepts, ideas, beliefs, or other items linked to, and arranged around, a central key phrase or idea [1][4]. These mind maps were useful to visualize, structure, and classify the beliefs and to organize related information. We guided a validation of the findings with the participants of the study in a focus group discussion. The group's responses and perceptions were analyzed both individually and in aggregate.

An important part of the mind map showed the *common beliefs* shared by the Product Owner and Scrum Master [15]. For example, both of them agreed that a good automated software testing approach is essential to the success of any project. The origin of this belief of the Product Owner was the experience in the current project. But, for the Scrum Master, the origin was in his past experiences in another company. They also cited some negative impacts on the project practices due to the absence of the automated software testing approach. Thus, the resulting mind map contains classes to represent the related elements: belief, meaning, origin and impact.

The study showed that the beliefs originated from two main sources: past and current project experience. Some beliefs triggered the adoption of new technologies, but the decision of which technology to adopt was mostly based on expert opinion and the current software engineering community buzz around it. We found *conflicting beliefs* between the Product Owner and the Scrum Master, which hindered the adoption of practices.

Some *conflicting beliefs* were subtle. For example, the interviewees agreed that current effort estimation practice was limited, but digging further, we found that they disagreed on which practice should substitute it. An even more subtle problem was when the interviewees had a *common belief*, but did not act upon it. We named those *semi-beliefs*. An example was the applicability of software testing automation for the organization; both Product Owner and Scrum Master seemed to believe in testing automation, but they were not adopting it.

Lastly, there were cases of common *strong beliefs*. This was the case of component reuse. Both, the Product Owner and Scrum Master actively acted on the direction of implementing good reuse practices from the very beginning of the project. This type of common *strong beliefs* were strengthened and transferred to other people involved in the project.

4. CHALLENGES OF APPLYING ETHNOGRAPHY

During the case study we identified five key challenges of using ethnographic methods to study software practices. Some of these challenges have also been discussed in the social science literature [8][11][12][14] and in SE literature [2][10][17][22]. We related our findings to relevant ethnographic theory, [3][7] [8][11][12][14] [24][25] to contribute to an improved understanding on how to apply ethnography to study software practices.

4.1 Collaborative Ethnography

A more cooperative and participatory perspective to ethnography is commonly required in organizational contexts. The challenge is to redirect and reinvent ethnography along such lines as collaborative ethnography – the collaboration of researchers and subjects in the production of ethnographic results, involving a side-by-side work of all parties in a mutually beneficial research program.

In Software Engineering is intrinsic in order to not only analyze, but also improve work practices. When doing ethnography through participant observation, the researcher must assume an attitude toward *being there* enough to experience the environment and nuanced aspects of socio-cultural life. This, at first, does not *allow* a collaborative approach to the research. Therefore, a researcher aiming to conduct ethnography faces the challenge of not influencing the practices in its natural setting.

In our case, at first we were only going to interview and observe the team members about their beliefs, but we noticed that we also needed to give feedback to the team about their common and conflicting beliefs, so that they could act as a unit and resolve the conflicts that hindered the adoption of some practices. We then ran a focus group together with a feedback meeting on the results of the study. As a result of this meeting, the team was able to prioritize action to solve the problems in the team. When confronting with conflicting team's beliefs and trying to find solutions for the conflicts, it was important that the team members knew that we were impartial towards the beliefs of how they were impacting the team practices. Considering our ethnographer role and required skill for the study, we sought to communicate well and openly, also being honest, trusting, realistic, and objective on our research.

Engaging in a shared process with the practitioners freed access to some more subtle dynamics of the team processes and practices. As participant observers and interviewers we realized, for example, that the Scrum Master and Product Owner were concerned about story and task estimation. At that time, the stories were estimated by hours and the Product Owner was responsible for this task. The Scrum master clearly disagreed with this strategy. He believed that the team should be more autonomous to estimate task and stories using the story points technique. The task estimation practice was, therefore, not working well for the project, so together we listed some actions to be done to improve it.

We dealt with this challenge in a similar manner as reported in social science literature. We have adapted our approach to ethnography grounded in collaborative research practice and we have engaged in mutual knowledge exchange with the participants, based on Lassister [11] and Lewis and Russell's [12].

A resembling approach was adopted in some other SE studies. Boden et al.'s study [2] introduced the concept of Business Ethnography and illustrated their experiences of adapting and implementing a collaborative approach using a typical action research cycle: research, analysis, and feedback workshops with the aim of finding better ways of cooperating across temporal, spatial and organizational barriers. Robinson et al. [17] used the findings to uncover and understand problems and thereby delineate their solution spaces. Their study revealed potential solutions to some problems related to knowledge sharing, using the interview responses to provide a context in which the problems could be elucidated and possible solutions could be discussed.

4.2 Insider/Outsider Dynamic

The participant observation technique also requires the ethnographer to have both, an *insider* and *outsider* posture [12]. The challenge is to combine participation with observation, seeking to understand the social reality and also trying to see familiar settings with a professional distance that allows adequate observation and recording of data [7]. There is a need for a balance in the researcher's participation in the team.

In our context, we were able to identify some common and conflicting beliefs in the team. Some conflicting beliefs were subtle and our approach of being inside their world at one moment and offering an outsider's perspective in another moment definitely helped to identify and point out those conflicts between the involved stakeholders. For example, during the observation of the Scrum meetings, we could perceive that some issues related to Scrum, such as self-management team, generated conflicting beliefs between team members and the Product Owner. We had the opportunity to confirm this during a subsequent interview. This finding was only possible because we assumed an insider position, so we were able to see the interactions and relationships between the people under observation with an inside perspective of context.

Sometimes the researcher can assume the role of being only an insider and this attitude may hinder some important discoveries. In the initial interviews we did not explore some possibilities related to participants' responses because we thought we already knew the answer for them. Later, we realized that this was a mistake, because we had not properly obtained the participants' point of view. In an attempt to mitigate this, we performed complementary interviews to ask more details about origins and impacts of beliefs related to some practices.

On the opposite end, when the researcher is only an outsider, some mistaken interpretations can arise because the researcher does not manage to take an appropriate insider posture. The drawbacks of this situation also involve difficulties to overcome data access barriers. In our case study, we had full access to recordings of the project meetings because the participants were comfortable to give us confidential material. As insiders, they knew we were able to handle it and to interpret the data properly.

In some previous studies [2][12][17], as well as in our study, the ethnographers needed to know how to stand in-between their research subjects, contexts, and demands, and the ethnographic

product. In our opinion, researchers must evolve their participant observation skills to accommodate an embedded analytic vision. In Boden et al [2], the researchers actively looked for a plurality of views to analyze the findings and their implications for the studied projects, taking into account their own roles in the projects, and constantly reflecting on them during the research, even in times when it was only possible to conduct remote interviews with team members of different sites. Lewis and Russell [12] identified and discussed the insider/outsider dynamic of participant observation as one of the key elements of embedded research practice. They conducted the research as some kind of team member and, even when they were not allowed to engage as a participant in clinical practice, for example, they did not stop to work alongside their co-workers in the community. Finally, Robinson et al [17] discussed their experiences and put the observations in the context of the study in order to maintain their perspective of strangeness and reduce the possibility of introducing bias.

4.3 Observation and Listening

Participant observation traditionally involves direct observation as the method of data collection. Other appropriate methods involve informal interviews, participation in the life of the group, collective discussions, analysis of personal documents produced within the group, self-analysis, results from activities undertaken off or online, and life stories.

Participant listening is an important technique employed by ethnographers, particularly among those who live in an interview society, where interviewing has become common practice. It is an appropriate way of participating and getting involved in software development contexts. The interviews allowed a deeper understanding of the obtained data through our observations [8].

In our context, beliefs are intangible. They are in the head of the team members. Usually beliefs are not externalized as part of their daily activities. It was very challenging to only observe the team members and try to understand what each person's beliefs were. We employed an interview-based qualitative data collection technique called *storytelling*, which adopts a specific story form, *the war story* [13]. Our war story questionnaire went beyond asking questions that allowed the team members to retell and revive specific and directed stories that illustrated the experiences we were trying to capture. So, it was possible to obtain personal opinions on the impacts of beliefs on agile team practices. Once we had identified the beliefs, we could observe their actions upon them.

For example, despite the fact that the Scrum Master and Product Owner shared the same belief of using tools for test automation, they did not adopt these tools in the project. We could observe that the Scrum Master and Product Owner did not seem to act according to their beliefs and, consequently, there were some negative impacts on project practices such as: no time left in sprint for bug correction; poor coverage and low effectiveness of the test and inspection practice; and greater effort to perform regression testing. In the interviews with the Scrum Master and Product Owner we identified that they did not really believe that tools for test automation were the appropriate solution for their test problems. Thus, the data collected in these interviews enabled connections between the participants' stories and our observations. Thus, we strongly recommend that SE researchers be both a participant listener and observer, because the data captured in the field can reflect more of what is heard than what is seen; casual

conversations and formal interviews can be construed as part of what is observed in the field.

The same approach is advocated by several authors who argue that the practice of participant listening should sit alongside with participant observation as an equally valid way of gaining ethnographic knowledge [2][8][16][17]. In order to get a complete picture, the researchers deploying Business Ethnography [2] relied on intensive, individual interviews with all participants involved in the project alongside other ethnographic methods like on-site observation to complement the interviews, which helped them to get a rich understanding of the project context that is related to the different perspectives, intentions, and expectations of the participants. Forsey's hypothesis [8], that ethnography is at least as much about conversation as it is about observation, was confirmed when his thesis and subsequent monograph was not devoid of observational description, but the data presented reflected more of what he heard in the field than what he saw. In Reeves et al.'s research [16], ethnographers routinely used informal or conversational interviews during their observations, which allowed them to discuss, probe emerging issues, or ask questions about unusual events in a naturalistic manner. To answer the first research question of the study related to influence of social factors, Robinson et al [17] collected data from organisations using a combination of semi-structured interviews and ethnographic-based observation.

4.4 Relationship with the Participants

The relationship between researchers and participants in an ethnographical work is a core element for successful research. As the researcher immerses in the organizational context, this relationship tends to grow and improve gradually, as people get to know each other and become closer, or they gradually deteriorate. People in a relationship tend to influence each other, share their thoughts, beliefs and feelings, and engage in activities together.

In an ethnographic study, the researcher becomes a member of the community under study, since the researcher has an active role in the target projects. In this context, he or she could reasonably be described as a practitioner and as a research scientist at the same time. But being a member of the community under study has both challenges and advantages for the researchers. The main advantage is that researchers and practitioners share the same organizational culture and so researchers are accepted and have common vocabulary with the participants, therefore the participants can relax and focus on working in their natural way. The challenge lies in the tension involved in moving between two roles (researcher and practitioner), and maintaining a non-judgmental orientation and a perspective of strangeness. At the same time, ethnographers have to be flexible, patient, and persistent in their work to overcome data access barriers.

In our study, we had to be non-judgmental in order to avoid bias in collecting and interpreting data. For example, during the focus group meeting two project team members discussed some points related to conflicting beliefs. For the Scrum Master the team should avoid direct contact with the customer, but for the Product Owner this direct contact between the team and the customer could impact positively the team's commitment to the quality of the product delivered and sense of business value. We had to keep a neutral position throughout this discussion. We also sought to minimize our effect on the behavior of the subjects under study, trying not to get over-involved with the people and environment to not disturb the natural setting.

In Boden et al.'s study, the researchers were subjected to similar challenges as the practitioners [2]. Known problems of conducting research on global software development teams included getting access to the research field in the first place as well as dealing with constraints such as organizational hierarchies and conflicting interests that might hinder a deeper immersion of the researchers in the environment [2]. Reeves et al.'s research described the relationship between researcher and the participants and the ethical issues, related to informed consent agreements, which arose from this close relation [16]. In Robinson et al.'s study, the relationship between the researchers and the participants was particularly significant because ethical issues concerning formal contracts or informed consent agreements were problematic in their experience [17].

4.5 Rigor and Contextualization

In the context of ethnographic work, the researcher is the main instrument of the ethnographic study. It is more of a general style rather than following specific prescriptions in a procedure [18]. Rigorousness in data collection and analysis is essential in order to avoid bias.

One aspect of the rigorous approach is that all findings must be grounded in the data and it is important to look for disconfirming instances, seeking behavioral confirmation, not just what people say but what they actually do. One challenge is to deal with a substantial amount of data derived from the ethnographic process. We used tools to deal with the collected data, which helped to work on the data in several iterations. We attended and recorded 40 meetings involving the interviewed participants and all the team members. For each hour of recorded interviews, we spent an average of five hours transcribing with an average of 13 pages, and one hour of pattern coding to tag key words, phrases, and paragraphs. The data collected was then analyzed via cycles of pattern coding which grouped segments of data into 26 themes.

In terms of validity of the results, we checked the answers from the interviews with our observations during the project. When the conclusions were reached, we discussed them with the team members in a focus group meeting. With this practice we prevented inappropriate assumptions or interpretations.

For us, theory and ethnography should be conjoined to produce a concrete sense of the social, so we conducted our ethnographic study with an underlying theory, building a conceptual framework to represent graphically the beliefs, their origins, sources, and relationships with team practices. This theoretical research approach helped to define the problem and to limit the research scope and effort, establishing a baseline to begin observations and the research over time.

In pursuing rigor in the research also involves the dilemma of sufficiently broad and detailed contextualization. Contextualizing data involves placing observations and interviews into a larger perspective. Each scene exists within a multilayered and interrelated context [7]. Contextualization helps to provide a more accurate characterization of the software practices and helps to prevent misunderstandings that can happen especially when a setting is familiar. One example is that sufficient description and quotations were included in our reports to allow readers to understand our research context. However, no study can capture an entire culture or group context; even a comprehensive report has to omit a great deal.

In our case study, the participation of the first author in the Scrum meetings of the project helped us to realize that there were

common strong beliefs, like component reuse, for example, in which both the Product Owner and Scrum master, actively acted on the direction of implementing good reuse practices from the very beginning of the project, even with several unfavourable conditions. Only with the contextual information about the origins and sources of their beliefs and the project environment was it possible to capture and understand how strong these beliefs really were.

In Robinson et al.'s study [17], rigorousness in data collection takes the form of three main principles: triangulation, seeking disconfirmation, and iterative development of understanding. The researchers employed triangulation by using data from different sources, for example, from interviews with different people having similar or different roles, from the artefacts, from observations of different roles and from informal conversations. They also focused on behavioural confirmation and searched for disconfirming instances during analysis and used both formal and informal feedback, presenting tentative findings to the participants.

5. DISCUSSION

This study highlights the methodological challenges involved in applying ethnography to study software practices. Here we discuss these challenges all together in order to support others who wish to conduct this type of study. Also in this section, limitations and implications for research will be presented.

5.1 Implications for Research in SE

Considering challenges 4.1 and 4.2, we highlight that it is not easy to immerse in a company with a dual objective of improving organizational problems and generating scientific knowledge. It demands an additional set of knowledge items and skills on the researcher in order to well conduct the process and provide relevant results. The researcher needs to formulate theories and ideas, prepare theoretical explanations, and establish collaboration with the people and the organization. In this scenario, it is important not to lose focus on the research goals and make constant reviews of the study plan and protocol. This is a product of the insider/outsider dynamic of participant observation, which requires the ethnographer to have both, an insider view as a practitioner and the ability of having an outsider's perspective as a researcher.

The study also shows a clear need for a particular way to deal with ethnographic research involving software development practices. SE contexts require a new approach that aims not only to analyze but also to improve work practices, engaging in a shared process with the practitioners. The researcher needs to demonstrate an ability to collaborate with the setting, understand the social reality and, at the same time, offer an outsider's perspective. Our study also highlights that it is important for the researchers to minimize their effects on the behavior of the people being studied. They should assume a collaborative posture and have something to offer to the participating company, but must avoid getting over-involved with its environment.

During the study, we made use of techniques that showed to be feasible, like a combination of interviews, observations, and document analysis. No doubt, the use of an interview technique contributed to meet challenge 4.3 through the balance of the practices of participant listening and participant observation, and the provision of a better perception of the participants' stories and their connection to the observations. The evidence suggests that researchers should consider the possibility of placing engaged

listening on a similar footing to participant observation. After some casual conversations and formal interviews, the big picture can be construed as part of what was observed in the field, so it is relevant to say that the data captured in the field can reflect more of what is heard than what is seen.

In ethnographic studies the research instrument (the ethnographer) needs to be properly calibrated to ensure the validity of the findings. The premise of *going native* cannot result in discontinuation of the study or moving from the role of researcher to the role of advocate. However, some conflicting interpretations among the participants and researchers can emerge. To avoid this situation and overcome challenges 4.4 and 4.5, we adopted the practice of discussing and validating field notes with the participants. The participant's perceptions were analyzed both individually and on an aggregated basis in order to build the big picture of the findings and their relative importance. As mentioned before, this practice contributes to rigor and can mitigate inappropriate interpretations.

Considering the ethnographer role and required skill for this type of research, we recommend that the ethnographers seek to communicate well and openly, participate fully, work together with the participants, and be honest, trusting, realistic, and objective. Also, fieldworkers have to be flexible, patient, and persistent in their work to overcome the inherent barriers and difficulties of data collection and analysis.

Our findings indicate that researchers should balance the role of participant observer with rigorous fieldwork. Rigorousness in data collection and analysis is essential in order to avoid bias. It is also important to look for disconfirming instances. Methodological triangulation is a well-suited approach for this purpose as it can be used to perform a cross-examination. By combining multiple observations, theories, interviews, and empirical materials, researchers can overcome the weaknesses and intrinsic biases, address issues of validity and problems that occur during ethnographic studies, and obtain comprehensive insights into the application of ethnographic methods.

The final product of the ethnographic study will depend directly on the decisions of the researcher. These decisions are crucial to allow the production of relevant scientific findings, but also to reconcile them with the organization's business needs and meet challenge 4.1. Here, again, careful planning and execution has to be considered. In spite of that, it is still important to be flexible and adapt to occurrences of the daily project activities.

5.2 Limitations

The general critics of a methodological study based on a single case study also apply to an ethnographic study, among them one may list: uniqueness, difficulty to generalize the results, and introduction of bias by researchers.

In our study, we transformed the findings from empirical statements to theoretical statements, which involve generalizing data from observations and perceptions by discussing them in accordance with the ethnographic literature. In this respect, we related our findings to relevant examples of ethnographic research within the SE literature and compared them with the related ethnography theory.

Another limitation is that we were working with the findings of one particular project within one particular organization. The participants were professionals using typical development technologies in a typical working environment, e.g., the natural

setting demanded by ethnographic approach. Nonetheless, the case study's environment is quite context specific, since the participants belong to the same company.

In addition, the study shows that running ethnographic research in SE has some specific limitations. It is not easy to get involved in a software company with a dual objective of solving organizational problems and generating scientific knowledge. In a competitive industry like SE, to get information on projects, processes, and practices is not an easy task, because of confidentiality issues and the fact that empirical research is not a high priority for this industry. Hence, in order to achieve a successful research project, a combination of ethnography with other strategies, such as action research, can provide relevant results and overcome some intrinsic limitations.

We should say that we do not have a complete list of challenges, thus, further studies should be performed to point to other challenges of applying ethnography in SE contexts. Also, there is a risk that our findings could be influenced by factors that escaped our attention. One common view is that it is a good practice to discuss and validate findings with other researchers and with the participants to seek the completeness of the conclusions. We have done this partially among the researchers involved in this paper.

6. CONCLUSION AND FURTHER WORK

Applying an ethnographic approach can be both challenging and demanding, it requires a long time to collect data, sometimes years. Hence, it becomes necessary to take into account the influence of the researchers' presence and their effects, regarding the aim to support and collaborate with the participating companies, rather than just studying them.

In this study we identified five categories of challenges of applying ethnography in a case study involving an agile software project. Each challenge was described in terms of the current state of the practice and how we conceived the goals for future works.

Several ways of addressing the challenges were outlined. Together, the challenges suggest that there is a particular way to deal with software development practices, which requires approaches that aim explicitly to construct a cooperative workspace, encompassing the researchers and the project team.

It is important to emphasize the different types of observer roles and the degree of participation in order to, in some moments, be an outsider and, in others, an insider. Besides that, we recommended to use methodological triangulation to avoid bias due the close relationship with the participants.

Our next step is to conduct new ethnographic studies with other software projects. Through the synthesis of the evidence, we intend to contribute to provide rich narrative accounts of the ethnographic research activity, and elucidate more questions and issues that arise from the use of ethnographic methods to study the practice of software development.

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APPENDIX A: INTERVIEW QUESTIONNAIRE

Warm-up questions:

1. What methodologies, software architectures, technologies, application domain, and types of client and size of projects have you worked with?
2. What are the main challenges of your current project?

Past experience questions:

3. Could you cite a past experience where the absence or presence of well-defined work process impacted (positively or negatively) the project's progress?
4. Could you cite a past experience where the absence or presence of risk and communication plans impacted (positively or negatively) the project's progress?
5. Could you cite a past experience where failures in software engineering practices impacted the project's progress?
6. Could you cite a past experience where the absence or presence of project monitoring process by metrics impacted (positively or negatively) the project's progress?

Lessons learned questions:

7. Could you tell me different and similar practices adopted in past projects and in the current project?
8. What best practices related to your expertise and experience were useful to apply to your current project? Could you tell me about their application?
9. What situation or risk have you tried to prevent or mitigate in the current project on account of experiences already lived? Cite and comment.

Reaction questions:

10. In what aspects is the software development methodology of the company impacting your project?
11. Have any unexpected effect or impact happened after your current project started to use this methodology? Cite and explain.

Metric-based questions:

12. What affects, positively and negatively, the productivity of your project? Cite and explain.
13. What affects, positively and negatively, the quality of your project? Cite and explain.
14. What affects, positively and negatively, the time schedule of your project? Cite and explain.
15. What affects, positively and negatively, the cost of your project? Cite and explain.

APPENDIX B: OBSERVATION INSTRUMENT

Specific Goals to Record:

- personal values, paradigms, and folktales of the project team members.
- interactions between project team members.
- activities actually performed by the project team.
- practices adopted by members of the project team.
- impact of the practices of team members on the project.
- arguments for and against a particular practice.
- influence of a practice on another practice in the project.
- events that happened during the work of project team.
- exceptions during the work of project team.
- tools used by the project team members.
- time spent on activities performed by project team.
- metrics that show the practices adopted by the team.
- other phenomena relevant to the research.

Guided questions:

1. What activities are actually performed by the project team?
2. What practices are actually adopted by the project team?
3. How often these practices are adopted by the project team?
4. Are these practices being adopted properly by the project team?
5. Which profile adopts certain practices within the project?
6. How to understand the folklore behind the practices in the project team?
7. How to understand the meaning of folklore for the team?
8. How to understand the origin of folklore in the team?
9. How to perceive the impact of folklore on the project?

10. How to understand the influence of folklore on other folklore within the project?
11. How to characterize the folklore behind the practices in the project team?
12. How to track the impact of folklore on the project? (history, timeline, context, etc.)
13. How to measure the impact of folklore on the project?

APPENDIX C: CONTEXT INSTRUMENT

Company Context: ID: Company Name: Background: Creation: Location: Segment: Services Offer: Annual Revenue: Number of Employees: Certifications: Standard Process:	Project Context: ID: Project Name: Main Goal: Domain Customer : Origin: Customer Type: Customer Segment: Start Date: Size (PF): Productivity(H/PF): Time: Technology Platform: Methodology: Tools: Metrics:
Project Team Context: Member Name: Role: Role Description: Seniority: Experience Time: Competences: Mini-CV:	